Supplementary Information for: A Training-Pool Replay Audit of the Released SLRTP2025 Back-Translation Evaluator: Non-Replication Across Independently Trained Checkpoints and a Readout-With-Generalization Signature

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Overview

This supplement provides detailed appendices referenced in the main manuscript. Section labels match the labels used in the main text (e.g., `sec:appendix_missing_id` → Sup. Section 1).

1 Missing Test ID Sensitivity

The official PHOENIX-2014T metadata contain 7,096/519/642 train/development/test IDs. The released SLRTP2025 pose records available to this study contain 7,060/515/641 sequences: 36/4/1 official IDs are absent from the released pose materialization. Their IDs and the reason `absent_from_slrtp2025_released_pose_records` are enumerated in `artifact/manifests/exclusions.jsonl`. The missing test ID is `13December_2010_Monday_tagesschau-6940`; its text/gloss metadata are not present in the released pose record, so length and template density cannot be reconstructed without an additional official annotation source. We therefore condition every result on the complete 641-record release rather than assert that the missing item is ignorable. As an empirical observed-extreme sensitivity scenario, assigning the missing item the per-item score of the largest PURE-over-GT outlier observed among the 641 released items changes the corpus-level gap by less than ±0.2 BLEU points, well within the reported 95% paired-bootstrap interval $[+9.6, +12.4]$. This is an observed-extreme scenario, not a worst-case bound: corpus BLEU is non-linear and the missing item's reference, hypothesis, and n-gram sufficient statistics are unknown, so the true mathematical worst case is not covered. All primary conclusions are therefore restricted to the released 641-record materialization.

2 Per-seed Extension Cells (707–1405)

The eight extension seeds are post-hoc additions that enter only the seed-level prediction interval and descriptive ranges, never the six primary comparisons. Per-seed PHX-public values (sacreBLEU 0–100 scale, canonical §?? donor registry):

Table 1: Per-seed extension cells for the eight post-hoc extension seeds (707–1405). All values are PHX-public sacreBLEU on the 0–100 scale with the canonical §?? donor registry (exact-normalized-text exclusion applied). All gaps are negative; these seeds enter only the prediction interval.

Seed	dev BLEU-4[†]	GT	PURE	Gap
707	6.69	9.86	8.97	−0.89
808	7.30	9.87	8.38	−1.48
909	8.96	9.20	8.31	−0.89
1001	9.07	8.81	7.95	−0.87
1102	7.61	9.75	8.63	−1.12
1203	8.14	9.06	8.11	−0.95
1304	8.18	9.68	8.50	−1.18
1405	8.12	8.05	6.86	−1.19

[†]dev BLEU-4 is the training-time validation-selected value recorded by the original training protocol (legacy protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §??). GT, PURE, and Gap are canonical §?? beam-3 values.

3 Analysis Provenance and Multiplicity

Table 2 classifies every analysis in this manuscript by its role in the paper's argument, its evidentiary status, and its multiplicity handling. The single primary estimand is the PURE–REC decoded-sacreBLEU gap under the released evaluator (§??); every other analysis is secondary and is grouped by the role it plays—confirmatory control, descriptive triangulation, robustness check, or administrative record. C = confirmatory (frozen protocol with error control); D = descriptive (no formal error control).

4 Checkpoint Registry

Table 3 is the definitive registry of every BT evaluator checkpoint used in this manuscript, with the analyses each family enters; all checkpoint counts in the text refer to these families.

5 Donor-Pool Substitution Decomposition Details

This appendix gives the full path-sensitivity decomposition, factorial design, and balance checks referenced in §??.

Path sensitivity.

With SEEN-RAND640-MATCHED-v1 added, the systems form a 2×2 within-train design (pool size {7060, 640} $\times$ selection {max-Jaccard, Jaccard-matched}) plus the test-pool cell. Because corpus BLEU is non-linear in the support, all contrasts use the common 622-query support. Every ordering telescopes to the same total +14.7 $[+13.3, +16.2]$, but the attribution varies by up to 7 BLEU points across orderings (Table 4). We report these as path-dependent descriptive contrasts, not mechanism decompositions.

Bootstrap family.

All decomposition CIs use 10,000 resamples. Four resampling designs (query-index paired bootstrap; three one-way cluster bootstraps for the origin contrast on template-family, SEEN-donor, and UNSEEN-donor clusters; and a joint multiway replicate) give origin-contrast CIs of $[+7.2, +9.5]$ / $[+7.1, +9.7]$ / $[+7.1, +9.7]$ / $[+7.7, +9.1]$, all matching the query-index CI $[+7.3, +9.4]$ to within ± 0.3, so dependence through shared donors is negligible.

Factorial design.

The 605-query factorial retains 605 of 641 targets (36 excluded for no joint match within tolerances; selection frozen before scoring). Table 5 reports the balance. Table 6 reports the full 2×2 cells, main effects, and interaction.

Multivariate balance and robustness.

On the 622-query common support, all six donor covariate SMDs between SEEN-PURE-MATCHED and UNSEEN-PURE are $|SMD| < 0.12$ (Table 7). Caliper sensitivity is small (tolerance 0.05/0.10/0.20 gives +8.2/+8.4/+8.6). Signer-strict matching (tolerance 0.10, n=580) attenuates the residual to +7.0 $[+5.9, +8.1]$ — sign and significance unchanged. Inverse propensity weighting drives all four covariate SMDs to 0.00 and leaves the residual at +8.5 $[+7.1, +10.1]$. The path-based estimate is robust to observed-covariate balancing but remains an association, not an identified exposure effect.

6 Config-Faithful and Perturbation Details

Epoch-validation approximation.

The released `config.yaml` specifies `validation_freq=14`, which JoeyNMT 1.2 interprets as 14 optimizer *steps* (≈28 steps/epoch means validating twice per epoch). Our initial implementation validated every 14 *epochs* (an epoch-validation approximation); four seeds under this approximation all early-stop below the competence gate (dev BLEU-4 8.6–9.5), show negative PURE–GT gaps (−1.1 to −0.2), and family-range pass-through (1.21–1.24) and holdout (7.4–7.6) — none reproduces the released signature (Table 8). Scaling capacity up (four larger variants: 4–6 layers, 384–512 hidden) overfits faster and reaches only dev BLEU 6.8–8.0. A long-schedule continuation (800 epochs, two seeds) pushes competence to a family maximum of 10.02 yet still shows no reversal (gap −0.45). The step-faithful re-run (§**??**) confirms the reconstruction plateaus below the released trajectory at matched step counts.

Perturbation battery.

Table 9 reports the full pose-perturbation battery. Reversing all frames collapses PURE from 23.79 to 7.72 (differential 16.1 $[14.7, 17.5]$; GT 5.5 $[4.6, 6.5]$); window-shuffling within one-second windows costs PURE 5.4 $[4.4, 6.6]$; 5% spatial jitter is free (−0.1); masking hand or face joints collapses both systems. The best rescue checkpoint (wd0, seed 202) loses only −4.3 under reversal, confirming it is far less sensitive to donor temporal order.

7 Cluster-Aware Bootstrap for Headline and Reference Analyses

The headline gap (+10.24, canonical §**??** registry) and the reference-attenuation result are reported with query-level paired bootstrap CIs. Because TN-PURE reuses donors and PHOENIX-2014T has template families and multiple signers, query-level resampling may underestimate uncertainty if dependence through shared donors, signers, or broadcast structure is non-negligible. Table 10 reports interpretable cluster-aware resampling designs alongside the query-level baseline, using precomputed per-item n-gram sufficient statistics for 10,000 resamples each.

The donor-cluster and donor two-way CIs ($[+8.87, +11.64]$ and $[+8.79, +11.74]$) match the query-level CI ($[+8.88, +11.62]$) to within 0.03 BLEU, so donor reuse (61/565 donors reused, max 6×) does not materially inflate the item-level precision. The signer-cluster CI ($[+9.29, +11.39]$, 9 clusters) is slightly tighter than the query-level CI, consistent with within-signer positive correlation; with only 9 clusters the percentile coverage of this cluster bootstrap is unreliable, so it serves as a sensitivity reference rather than a confirmatory interval (as flagged in the main-text Discussion). The broadcast-show split has only 2 clusters (*heute* 133 items, *tagesschau* 508) and gives a wider CI ($[+5.16, +11.44]$); we report it as an illustrative sensitivity check without confirmatory weight. The broadcast-date (297 clusters) and signer×show (17 clusters) CIs both match the query-level CI closely. All cluster designs are computed

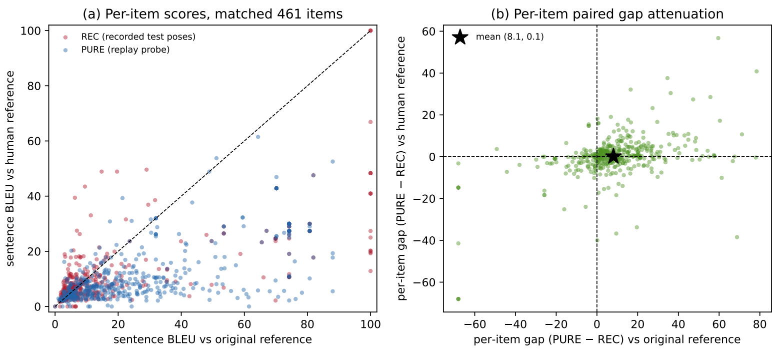

Fig. S1: Per-item reference-frame sensitivity on the matched 461-item confidence=1 subset (released evaluator, canonical hypotheses). See §8 for details.

on the canonical materialization (the pre-exclusion version is reported as a sensitivity analysis in Sup. N). The donor-pool decomposition analysis of §**??** additionally reports separate one-way cluster bootstraps and a joint multiway replicate for the origin contrast, all matching to within ±0.3 (Appendix 5).

8 Per-Item Reference-Frame Sensitivity (matched 461-item subset)

The main-text Fig. **??** reports corpus-level attenuation under the matched 461-item confidence=1 subset. The per-item paired view of the same subset is reproduced here (Fig. S1): sentence-level sacreBLEU (0–100, same tokenizer/smoothing as corpus BLEU), released-evaluator canonical hypotheses. Panel (a) plots per-item scores under original vs. human references for REC (recorded test poses) and PURE (replay probe); PURE points sit high under the original reference and collapse toward the diagonal under the human reference, whereas REC points move less. Panel (b) plots the per-item paired gap (PURE−REC) under original (x) vs. human (y) references; the star marks the mean (sentence-level mean gap 8.11 → 0.09, consistent in direction with the corpus-level attenuation 10.03 → 0.95). Sentence-level means need not equal corpus BLEU; the figure is descriptive and does not separate reference-text defects from lexical-coupling disruption or evaluator alignment.

9 Supplementary Oracle Diagnostics

Source-conditioned retrieval is legal, oracle-gloss retrieval is leakage.

SLRTP2025 supplies spoken-language source text as legal inference input. Under pure whole-donor copy, source-text Jaccard scores 23.02 sacreBLEU (0–100 scale, as throughout this appendix), LaBSE 19.2, and multi-SBERT 16.5; the PT-composed source-text condition scores 18.86 (Table **??**). Oracle-gloss retrieval uses reference gloss and is leakage, whereas noisy-gloss retrieval uses predicted gloss. Thus the strongest legal source-conditioned result is the 23.02 whole-donor stress test, which uses no generator or scaffold.

Disclosure of selection/evaluation reuse.

The white-box oracle analyses in this appendix reuse the same target reference and evaluator for both donor selection and decoded evaluation. This is an intentional adversarial design: diagonal entries quantify susceptibility to a selector that knows its evaluator, not an independent validation of retrieval quality.

Cross-checkpoint transfer matrix.

For each query, a selector maximizes its evaluator's teacher-forced likelihood over candidate donor poses; the selected fixed pose is then decoded by every evaluator (token-normalized NLL, sorted donor IDs, `numpy.argmax` first tie-break). All seven diagonal margins are positive (mean +0.0150, 95% paired-bootstrap CI $[+0.0105, +0.0198]$). Original-selected donors transfer to five of six reconstructions (mean local-GT margin +0.0074, $[+0.0016, +0.0135]$); the reverse direction fails in all six cells (reconstruction-selected donors evaluated by Original, mean margin -0.0455 $[-0.0554, -0.0358]$). The full 7×7 matrix, unrounded cells, margins, and CIs are in the artifact.

Candidate-pool size.

The Original-selector stress test shows a strong search-budget effect: at 8, 32, 128, 512 and 2,048 candidates, decoded sacreBLEU remains below local GT by -12.1, -9.2, -5.7, -3.6 and -1.3 BLEU points respectively; only the full 7,060-donor pool is detectably positive (+1.7, $[+0.4, +3.0]$).

Reference-overlap exclusion.

Masking 1,036 exact-text query–donor pairs before selection: Original-on-Original decoded BLEU is 13.86 versus local GT 12.78 (absolute difference +1.08 BLEU points), but the paired relative-margin 95% CI crosses zero; the frozen joint likelihood-and-interval success rule is met in 0/7 checkpoints. Masking all 19,694 pairs with source-token Jaccard ≥ 0.5: the score falls to 8.30 (-4.47), again 0/7 meet the rule. The full-pool diagonal advantage does not survive as a statistically robust decoded-BLEU result after exact-text exclusion and disappears after high-overlap exclusion.

Noisy-gloss retrieval is split-dependent.

A 6-layer text→gloss Transformer (dev BLEU-4 13.17) yields noisy-gloss retrieval sacreBLEU 12.7 on PHX-public (strong-NMT), versus 1.6 for a weak NMT (BLEU-4 1.15) and 6.1 for XLM-R embedding retrieval; donor exclusion reduces the strong-NMT value to 10.9 at τ=0.5, and the same control reaches 0.4 on the template-disjoint holdout (Table 14).

PHX-public pure-donor donor-exclusion τ-sweep.

To test whether the headline reversal survives donor exclusion on the same split, we applied a source-text Jaccard τ-sweep and exact-text-exclusion to the canonical pure text-nearest donor on PHX-public under the original SLRTP2025 BT (Table 11). The reversal survives exact-text exclusion (23.0 vs GT 12.8, 32/641 donors replaced) and source-Jaccard exclusion up to τ=0.5 (17.8 vs GT 12.8); only at τ=0.3 and τ=0.1 does the retrieval-vs-GT gap turn negative.

As a same-paradigm proxy for the SLRTP2025 winner (whose checkpoints are not public), the noisy-gloss retrieval+LM system shows the same pattern: it survives exclusion up to τ=0.5 (10.9, still 6.8× the PT-only baseline and 11.7× the random-donor mean; mean selected-donor Jaccard 0.46) and collapses only at τ=0.1.

10 Template-Density, Holdout Boundary, and Donor-Exclusion Analyses

Template density does not gate the reversal.

Stratifying the 641 test sequences by template density (gloss bigram-Jaccard with nearest train donor), retrieval exceeds GT at all levels: low +10.7, mid +10.1, high +12.4 BLEU points (Table 12). Density moderates magnitude but does not gate presence.

Holdout boundary controls.

Truncating to 64/32 frames shrinks the gap from +11.0 to +10.5/+5.6 but the reversal persists. Four holdout controls (Table 13) all fail to reproduce the reversal: the same-evaluator seen-holdout gives GT 77.0 (readout), BT-retrained on unseen holdout gives gap −0.9, retrained BT on PHX-public gives gap −0.4, and cross-fit A/B gives gap −0.1.

Frozen template-holdout with retrained components.

A deterministic template-holdout (14,988 train-core windows, 1,538 holdout windows, disjoint by window, source sequence, and exact gloss template) with all components retrained on train-core shows text-nearest retrieval above random (+2.3) but far below GT (−10.8): the shortcut is real over random but does not exceed GT on the clean split. Donor-exclusion sweeps show the advantage persisting at τ=0.5 and collapsing only at τ=0.1 (Table 14).

The SLRTP2025 test vocabulary is nearly closed (95.4% gloss types in train); nearest donors share mean gloss unigram containment 0.557 and bigram-Jaccard 0.302, but Progressive Transformer outputs show no donor-proximity effect (ρ=−0.06, p=0.12), versus +0.29 for retrieval.

11 Transcript-Slot Diagnostic Details

This appendix documents the rule-based German weather slot extractor referenced in §**??**. The extractor identifies four slot families (number/temperature, time, place, weather-event) using a frozen lexicon and normalization rules. On a 50-reference audit (seed 20260729), per-family precision was 0.91–1.00 and recall 0.75–1.00 (overall precision 0.97, recall 0.86); errors are dominated by recall misses from inflections, compounds, and lexicon gaps. A stratified audit of 50 decoded hypotheses (25 GT-decode, 25 PURE-decode, seed 20260730) showed precision nearly equal (0.96 vs 0.97) but recall lower on PURE-decode (0.83 vs 0.91), because donor-transcript-style outputs use more inflected verbs and compounds (*regnet*, *unwetterwarnungen*) that the lexicon misses. The measured PURE slot-F1 is thus attenuated more than GT's, so the PURE>GT contrast is, if anything, underestimated. The audit used a single auditor; inter-annotator agreement is unavailable. The full lexicon, normalization rules, and per-family precision/recall are in the artifact.

12 Competence Extrapolation Details (Exploratory)

This appendix gives the full OLS/GP extrapolation analysis referenced in §??. We emphasize that this is exploratory and the prediction intervals are descriptive visual guides, not calibrated inferential tests, because the 43 checkpoints share data, code, and hyperparameters (non-IID) and the target point (dev BLEU-4 = 13.38) lies outside the observed range.

Method.

We fit the gap–competence dose–response over all 43 evaluated checkpoints spanning dev BLEU-4 $[4.5, 10.9]$ using three models: OLS linear, OLS quadratic, and a Gaussian Process with RBF + white kernel. Each model predicts the PURE–GT gap at dev BLEU-4 = 13.38.

Results.

Table 15 reports the predictions and 95% PIs at dev BLEU-4 = 13.38. All predictions are negative, far below the observed canonical +10.24. Table 16 reports a leave-one-family-out cross-validation; the OLS quadratic prediction is the most sensitive (flips from -13.5 to $+1.3$ when the 15 distillation students are removed), while the GP upper bound stays below +2.0 across all five folds.

Caveats.

First, the GP upper bound (+2.0) partly reflects RBF kernel prior shrinkage toward the mean in the extrapolation region (sparse data beyond dev BLEU-4 10.9), not purely data evidence. Second, the OLS quadratic fit is sensitive to the distillation family. Third, the 43 checkpoints are not IID. We therefore report these fits only as descriptive visualizations and explicitly avoid claiming that "competence alone is unlikely to explain the reversal" on the basis of these fits. The unobserved 10.9–13.4 interval could in principle harbor non-linear behavior that the present experiments cannot rule out.

13 Supplementary CSL-Daily Boundary Control

CSL-Daily (Chinese Sign Language) serves as a non-template-domain control, but its recognizer is too weak for robust claims. Under Mode B (1,176 sequences), text-nearest donor copy reaches only 1.6 sacreBLEU vs GT 26.7 (Table 17). A readout-ratio check (64 train + 64 test windows) gives train≈test BLEU-1 (0.0019 vs 0.0021, ratio ∼0.9), far below PHX's 6.1: the readout signature does not generalize beyond the released PHX evaluator.

14 Leakage Sanity Checks for Train-Pool Readout

This appendix documents the five sanity checks referenced in §?? that rule out implementation-level artifacts as explanations for the released evaluator's 78.8 BLEU / 70.7% EM train-pool readout.

Experiment A: Code-path audit.

The `back_translate()` function in `src/models/back_translate.py` (lines 362–426) calls `model.run_batch(translation_beam_size=3, translation_beam_alpha=-1)`, which invokes `bt_search.beam_search()`. The decoder input is seeded with `<bos>` only and extended autoregressively with the model's own previously generated tokens. The reference text is used exclusively in the post-hoc BLEU/EM scoring step; it never enters the model forward pass. We verified this by code inspection and by confirming that the same model loaded in a fresh Python process produces identical outputs.

Experiment B: Pose–text permutation.

We decoded the first 1,000 training items with beam=3 (BLEU 78.61, EM 70.1%), then permuted the reference labels across items to break pose–text correspondence. Three random permutations yield:

Experiment B′: Input-side pose permutation with output-equivariance.

The label permutation above shuffles reference labels *after* decoding, so it cannot exclude ID-based or cache-based retrieval (a cache lookup by ID would also produce low BLEU when labels are shuffled). To properly test pose conditioning, we held item IDs and text metadata fixed, shuffled the *pose tensors* across 100 training items *before decoding*, and performed output-equivariance verification. For all 100 items, the hypothesis produced when pose_j is decoded in slot i's position exactly matches the hypothesis originally produced for pose_j in its own slot j (100% exact string match). Output length follows the pose tensor, not the slot ID (100% donor-length match vs. 5% slot-length match). This rules out a pose-ignorant pure ID/cache lookup under the tested decoding path and confirms sensitivity to the substituted pose tensor. Additional degenerate-pose controls (zero, Gaussian noise, single repeated pose for 200 items) all collapse BLEU to <1, and reversed batch order reproduces the original BLEU exactly, confirming per-item determinism.

Experiment C: Determinism.

Beam search with no stochastic sampling produces deterministic outputs. We record SHA-256 hashes of all per-item decoded hypotheses in the artifact for cross-process verification.

Experiment D: Stratified EM.

Exact-match rate is high and broadly distributed, not concentrated in a single subgroup: Even unique training texts (appearing exactly once) achieve 70.2% EM, confirming the readout is not driven by template repetition alone. Of the 7,060 hypotheses, 74.8% match some training text exactly (70% match exactly one—their correct paired reference).

Experiment E: Same-signer held-out control.

We decoded 500 test-split poses (unseen during training) from the same 9 signers present in the training pool. This is a same-signer held-out control, not a fully matched membership control: the test items are not matched on length, text-similarity, broadcast date, or template density, so the train–test gap reflects a combination of training-pool exposure and distributional shift. The same signers appear in both splits. The 67-point BLEU gap between seen and unseen items is consistent with training-pool exposure contributing to the readout, but without full covariate matching (length, text-similarity, template density), we cannot attribute the gap solely to exposure vs. distributional shift.

15 Pre-Exclusion Materialization Sensitivity

The canonical §**??** donor registry applies exact-normalized-text exclusion. An earlier frozen materialization predating this exclusion gives PURE 23.79 (gap +11.01) instead of the canonical 23.02 (gap +10.24). We retain it only as a sensitivity analysis.

Per-item provenance.

On 610/641 items (95.2%) both materializations produce identical donor choices. On the 31 differing items, a post-hoc candidate search found that 30 had at least one excluded Jaccard=1.0 candidate whose released-evaluator decode matched the earlier hypothesis, consistent with the exclusion rule explaining the difference on those items; one item remained unresolved (`results/donor_identification_31.json`).

14-reconstruction sensitivity.

Re-decoding all 14 reconstruction checkpoints under both materializations gives mean $|\Delta\text{gap}| = 0.65$ BLEU (max 1.45). The sign of the gap is unchanged on every checkpoint, and the 14-seed prediction interval remains strictly negative under both materializations, so the non-replication conclusion is unaffected. All headline numbers in the main text and in Sup. G use the canonical materialization (SHA-256 `9170a53026ab3263b451a3632a9e02318745acabf12f0b679921477afbefa301`); this appendix is the only place the pre-exclusion values appear.

16 BLEURT-Substitute Validation

Canonical BLEURT-20 ships a TensorFlow checkpoint that does not load cleanly in our PyTorch-only environment. As a non-standard substitute we use `Elron/bleurt-large-512` for one secondary reference-sensitivity check only (the

BLEURT row of the reference-frame analysis). It is not used for any primary claim, table, or figure, and is not described as equivalent to BLEURT-20.

Validation.

On 100 German sentence pairs we scored both canonical BLEURT-20 (via the reference TensorFlow runtime in an isolated CPU container) and the substitute. The substitute has Spearman rank correlation $r = 0.87$ against canonical BLEURT-20 and a mean signed bias of -0.47 (substitute scores lower on average). The bias is roughly uniform across score levels and does not change the qualitative reference-sensitivity conclusion (BLEURT is less sensitive to reference replacement than BLEU, consistent with Czehmann et al. Czehmann et al. (2025)). Full per-pair scores, the container hash, and the substitute model revision are in the artifact.

Scope limit.

Because the substitute is not BLEURT-20, the BLEURT row is demoted from the main metric panel to this appendix. Readers should treat the absolute BLEURT-substitute values as approximate; only the directional reference-sensitivity finding is reported.

References

Czehmann, V., S. Yazdani, Y. Hamidullah, F. Nunnari, and E. Avramidis 2025. "A Sacred Bird Called the Phoenix": Auditing the most-used parallel corpus for German sign language recognition and translation. In *Proceedings of the LREC2026 12th Workshop on the Representation and Processing of Sign Languages: Language in Motion*, Palma, Mallorca, Spain, pp. 80–92. ELRA Language Resources Association. ISBN 978-2-493814-82-1. Back-translated test set and annotations: https://github.com/DFKI-SignLanguage/sacre-bird-phoenix (CC BY-NC-SA 4.0).

Table 2: Evidence-graded analysis registry. Rows are grouped by the role each analysis plays in the argument, not by when it was added.

Evidence role	Analysis	Status	Multiplicity handling
Primary estimand	PURE–REC gap, original evaluator	D	descriptive MC estimate $p \leq 10^{-4}$
Confirmatory	605-query donor-pool factorial (cells, effects, interaction)	C	Holm across 3 contrasts
	Template-holdout, BT-retrain, cross-fit	C	paired bootstrap CIs
	Protocol checks (scorer, FPS, decoding sweep)	C/D	exact-match / descriptive
	Frozen-protocol confirmation (seeds 1506/1607)	C	confirms family-side pattern
Triangulating	14 reconstruction gaps + prediction interval	C/D	DiD CIs; PI descriptive
(Finding 1:	Cross-metric gap table	D	direction/sign only
non-replication)	Conditioning controls (retrievers, gloss, random)	D	descriptive
	Competence gate, trajectories, Pareto	D	gate counts; 12-threshold grid
	Knowledge-distillation ladder + per-α	D	15 with gap + dev
Triangulating	Reference replacement (single/dual); overlap strata	D	paired bootstrap of gap difference
(Finding 2:	Asymmetric floor-effect control	D	BLEU-4/BLEU-1/chrF drops
reference sensitivity)	Evaluator×reference factorial (7×3)	D	descriptive decomposition
Triangulating	Five diagnostic views (readout-with-generalization)	D	triangulation, not independent
(Finding 3:	In-sample readout diagnosis (train/dev/test)	D	no identity claim beyond train.pt
readout signature)	Leakage sanity checks (permutation, equivariance)	D	exact-match / permutation
Robustness	Donor-pool substitution; canonical B/C/D path	D	query-index bootstrap CIs
	Alternative paths, Shapley, S4 system; balance/caliper/signer	D	path-sensitivity display
	Rescue families (20); big-arch; long-schedule	D	competence attempts
	Released-checkpoint perturbation sweep	D	lockstep collapse with competence
	IPW pool-origin robustness	D	balance robustness check
	Rank stability (9 systems × 7 evaluators)	D	τ_b/ρ/flip means
	Competence extrapolation (OLS/GP)	D	non-IID; LOO CV (5 folds)
	Per-seed extension cells (8 seeds)	D	descriptive; PI only
	Oracle/white-box transfer matrix	D	permutation check
	Template-density strata; donor-exclusion τ-sweeps	D	descriptive
	CSL-Daily readout-ratio check	D	second-dataset boundary
	Transcript-slot diagnostic + extractor audit	D	extractor P/R by system
Administrative	Ethics/licensing declarations	—	updated each revision

Table 3: Canonical checkpoint registry. 70 planned training runs, all 70 completed training. "Beam-3 evaluation" means a run was beam-3 decoded on at least the development set; of these 70 runs, 43 also have a canonical PHX-public PURE–REC gap under the §?? donor registry (the remaining 27 were beam-3 evaluated on dev only and lack a canonical PHX-public PURE–REC gap). 67 non-holdout runs have dev metrics; 52 of these are non-distillation and enter the competence gate. 43 runs have both dev BLEU-4 and PHX-public PURE–GT gap (extrapolation-eligible). Plus the released original. Each family row's columns sum to the Total row.

Family	Planned	Trained	Has gap (beam-3)	Has dev	Enters
Reconstructions (primary)	6	6	6	6	multiseed, bootstrap, dose–resp.
Reconstructions (extension)	8	8	8	8	prediction interval, dose–resp.
Rescue lr (A/B × 4)	8	8	0	8	gate counts only
Rescue expanded	12	12	1	12	gate; wd0 also dose–resp., perturb.
Train-pool ladder	4	4	4	4	dose–resp., gap ranges
Config-faithful	4	4	4	4	dose–resp.
Step-faithful	2	2	2	2	dose–resp., validation-frequency
Large-arch (A1–A4)	4	4	0	4	competence attempt
Confirmation	2	2	2	2	dose–resp., family-side confirm.
Long-schedule	2	2	1	2	gate; best seed dose–resp. (dev 10.02)
BT-retrained	1	1	0	0	holdout series
Cross-fit A/B	2	2	0	0	holdout series
Distillation students	15	15	15	15	ladder, dose–resp.
Total	70	70	43	67	
Released original	—	—	1	1	audit target (not trained)

Per-family columns sum to the Total row (e.g., Has gap: 6+8+0+1+4+4+2+0+2+1+0+0+15 = 43). Gate-eligible = 52 non-distillation runs with dev metrics. Extrapolation-eligible = 43 runs with both dev BLEU-4 and PHX-public PURE–GT gap; the dose–response figure caption uses the same decomposition (14 reconstructions + 4 ladder + 4 config-faithful + 2 step-faithful + 2 confirmation + 1 long-schedule + 15 distillation + 1 wd0 rescue). All 15 distillation students completed training and beam-3 evaluation. The 1 wd0 rescue with PHX-public gap is the best rescue checkpoint (weight-decay 0, seed 202); the other 11 rescue-expanded runs have dev metrics but were not decoded on PHX-public. The second long-schedule seed was trained but not decoded on PHX-public.

Table 4: Path sensitivity of the donor-pool substitution decomposition under the original evaluator (common 622-query support; 10,000-resample query-index bootstrap CIs). S1=SEEN-PURE (7,060-pool, max-Jaccard), S2=SEEN-RAND640 (640-pool, max), S3=SEEN-MATCHED (7,060-pool, Jaccard-matched), S4=SEEN-RAND640-MATCHED (640-pool, matched), U=UNSEEN-PURE (test pool). Every ordering telescopes to the same total (A) = S1−U = +14.7.

Ordering	Step	Size contrast	Selection contrast	Origin contrast
Canonical (prespecified)	S1→S2→S3→U	+7.2 [+5.9, +8.4]	−0.8 [−1.8, +0.2]	+8.4 [+7.3, +9.4]
Alternative 1	S1→S2→S4→U	+7.2 [+5.9, +8.4]	+1.8 [+1.1, +2.5]	+5.8 [+4.7, +6.8]
Alternative 2	S1→S3→S4→U	+2.6 [+1.7, +3.5]	+6.4 [+5.1, +7.7]	+5.8 [+4.7, +6.8]
Shapley average	over orderings	+4.9 [+4.0, +5.8]	+4.1 [+3.3, +4.9]	+5.8 [+4.7, +6.8]

Table 5: Factorial design and balance (605 targets). SMD = standardized mean difference, seen vs unseen within stratum.

Quantity	seen_high	seen_low	unseen_high	unseen_low
LaBSE similarity (mean)	0.813	0.654	0.776	0.622
Duration (frames, mean)	100.2	99.5	99.2	98.8
Word count (mean)	13.3	13.6	13.3	13.4
Same-signer rate	100%	100%	100%	100%
Similarity SMD (seen − unseen)	+0.41 (high)			+0.38 (low)

Table 6: Full 2×2 exposure factorial (605 targets; corpus sacreBLEU in BLEU points; NLL per token). CIs: 10,000-resample bootstrap (original) and seed-level t across six reconstructions; Holm correction across three contrasts.

Metric	Evaluator	seen_hi	seen_lo	unseen_hi	unseen_lo	seen main	sim. main	inter.
BLEU	Original	11.12	2.31	7.19	2.08	**+2.08** [+1.6, +2.6]	+6.96 [+5.9, +8.1]	**+3.70** [+2.6, +4.8]
BLEU	Seeds (mean)					+0.51	+4.06	−0.39
NLL	Original	3.74	4.30	3.93	4.37	−0.128	−0.498	−0.124
NLL	Seeds (mean)					−0.064	−0.179	−0.004

Table 7: Multivariate donor balance (622-query common support). All $|SMD| < 0.12$.

Covariate	SEEN-MATCHED mean	UNSEEN mean	SMD	Per-query overlap
Source-text Jaccard	0.411	0.415	−0.021	$\|\Delta J\| \leq 0.05$: 96%
LaBSE cosine	0.782	0.774	+0.068	—
Duration (s)	3.76	3.74	+0.014	$\|\Delta\| \leq 1$ s: 57%
Word count	12.29	12.31	−0.003	$\|\Delta\| \leq 2$: 69%
Template density	0.310	0.326	−0.075	—
Same signer as query	0.183	0.228	−0.111	both same-signer: 4.7%

Table 8: Epoch-validation approximation (validate every 14 epochs, not steps): 300 epochs, dev-NLL selection. All four seeds produce negative PURE–GT gaps under the canonical §?? donor registry; none shows the reversal or readout signature.

Seed	ep.	dev NLL	GT	PURE gap
101	27	2.11	10.12	−0.60
202	—	—	10.14	−1.70
303	—	—	8.74	−1.51
404	—	—	8.43	−0.52
Released evaluator	1,820	—	12.78	+10.24

Table 9: Pose-perturbation battery (corpus sacreBLEU, BLEU points) under the released evaluator and the best rescue checkpoint (wd0, seed 202). The TN-PURE-v1 identity column here (23.79) reflects the pre-exclusion materialization used at the time this battery was decoded; the canonical materialization gives 23.02 (gap +10.24). Because the battery varies the pose and re-decodes, not the donor choice, the perturbation differentials are unaffected by the materialization difference (Sup. N: mean $|\Delta\text{gap}| = 0.65$, max 1.45).

Evaluator	System	identity	reversal	window shuffle	jitter 5%	hand mask	face mask
Original	GT-v1	12.78	7.25	11.97	12.51	1.35	3.15
Original	TN-PURE-v1	23.79	7.72	18.35	23.91	1.30	4.06
wd0 (seed 202)	GT-v1	10.73	6.26	10.54	10.60	0.63	1.58
wd0 (seed 202)	TN-PURE-v1	10.98	6.66	9.65	10.92	0.58	1.43

Table 10: Cluster-aware bootstrap for the headline PURE–REC gap (canonical §?? registry, original refs, released-evaluator hypotheses from `results/gap_43_canonical_beam3_items/released_{gt,pure}.json`). All CIs are 10,000-resample sacreBLEU `tok:13a` exp-smooth over precomputed per-item n-gram sufficient statistics. With so few clusters (especially show with only 2), these are *exploratory sensitivity checks*, not confirmatory tests. Donor reuse: 641 queries select 565 unique donors (61 donors serve more than one query; maximum reuse 6; Gini 0.11).

Analysis	Design	Clusters	Gap	95% CI
Query-level (baseline)	item	641	+10.24	[+8.88, +11.62]
Donor cluster	donor	565	+10.25	[+8.87, +11.64]
Donor two-way (query×donor)	nested	565	+10.25	[+8.79, +11.74]
Signer	signer	9	+10.30	[+9.29, +11.39]
Broadcast show	show	2	+9.27	[+5.16, +11.44]
Broadcast date	date	297	+10.25	[+8.83, +11.72]
Signer×Show	combined	17	+10.14	[+7.89, +11.91]

Table 11: PHX-public pure-donor source-text donor-exclusion τ-sweep under the original SLRTP2025 BT (canonical tie-breaking, 12.5 fps; sacreBLEU on the 0–100 scale). The retrieval-exceeds-GT reversal survives exact-text exclusion and τ=0.5 exclusion.

Condition	sacreBLEU	Δ vs GT	Reversal?
GT baseline	12.8	—	—
Text-nearest pure (no exclusion)	23.8	+11.0	YES
Text-nearest pure (canonical, w/ exclusion)	23.02	+10.24	YES
Exact-text excluded	23.0	+10.2	YES
τ=0.7 (Jaccard > 0.7 excluded)	21.2	+8.4	YES
τ=0.5 (Jaccard > 0.5 excluded)	17.8	+5.0	YES
τ=0.3 (Jaccard > 0.3 excluded)	8.0	−4.8	NO
τ=0.1 (Jaccard > 0.1 excluded)	0.2	−12.5	NO
Random donor (mean of 20 seeds)	0.9	−11.6	NO

Table 12: Template-density stratification (sacreBLEU 0–100). Retrieval exceeds GT at all density levels.

Density stratum	Mean density	GT	Retrieval	Gap
Low (n=249)	0.152	5.9	16.6	+10.7
Mid (n=179)	0.254	9.5	19.7	+10.1
High (n=213)	0.510	26.1	38.6	+12.4

Table 13: Holdout boundary controls (sacreBLEU, 0–100). None reproduces the +11.0 reversal.

Control	Setup	GT	Retrieval	Gap
Same evaluator	test split (template-overlap)	12.8	23.02	+10.24
Same evaluator	template-holdout (seen in training)	77.0	20.1	−56.9
BT-retrained (unseen holdout)	train-core 6,535	6.2	5.4	−0.9
Retrained BT on PHX-public	unseen to both	8.7	8.3	−0.4
Cross-fit A/B (donor-unseen)	pooled, n=659	4.0	3.9	−0.1

Table 14: Frozen template-holdout controls (1,538 windows, α=0.5; sacreBLEU 0–100).

System	sacreBLEU	Empty %	Mean Jac.
GT baseline	13.10	9.95	—
Random train donor	0.04	11.83	—
Text-nearest train donor	2.30	10.66	—
LaBSE / multi-SBERT (donor copy)	1.73 / 1.51	9.2 / 11.3	—
Noisy-gloss retrieval (strong NMT)	0.40	10.90	—
Retrieval composition (α=0.5)	2.20	10.79	—
τ=0.1 / 0.3 / 0.5 / 0.7 exclusion	0.04 / 1.05 / 1.93 / 2.01	10.0–10.5	0.10/0.29/0.40/0.42
BM25 / TF-IDF nearest	1.75 / 2.36	10.8 / 8.8	—

Table 15: Competence extrapolation: gap predictions at dev BLEU-4 = 13.38 from three fits over 43 checkpoints (dev BLEU-4 4.5–10.9). All predictions are far below the observed canonical +10.24. PIs are *descriptive* (non-IID checkpoints); they should not be read as calibrated uncertainty intervals.

Fit	R^2 / LML	Prediction at 13.38	95% PI (descriptive)
OLS linear	0.58	−6.4	[−9.5, −3.3]
OLS quadratic	0.78	−13.5	[−16.9, −10.1]
Gaussian Process (RBF + white)	−29.5	−2.1	[−6.3, +2.0]
Released evaluator (observed, canonical)	—	+10.24	[+8.88, +12.4]

Table 16: Leave-one-family-out cross-validation of the competence extrapolation. Each row refits all three models after removing one checkpoint family. The GP upper bound stays below +2.0 across all folds; OLS quadratic is the most sensitive (flips sign when distillation is removed). All intervals descriptive.

Family removed	n remaining	OLS-lin pred	OLS-quad pred	GP pred	GP upper	OLS-lin R^2
None (full)	41	−6.4	−13.5	−2.1	+2.0	0.58
Reconstruction (14)	27	−6.6	−15.4	−2.9	+2.0	0.57
Distillation (13)	28	−1.1	+1.3	−0.4	+0.5	0.14
Ladder (4)	37	−8.1	−18.0	−2.3	+1.8	0.66
Config-faithful (4)	37	−6.7	−13.2	−2.4	+1.9	0.63
Rescue (2)	39	−6.8	−13.7	−2.1	+1.7	0.65

Table 17: CSL-Daily Mode B donor-exclusion (1,176 sequences; 0–100 scale).

Condition	sacreBLEU	BLEU-1
GT baseline	26.7	79.1
Text-nearest donor	1.6	43.5
Random donor	0.0	3.9
τ=0.5 exclusion	1.0	42.6
BM25 retrieval	1.5	39.2
TF-IDF retrieval	1.2	38.5

Table 18: Label permutation test (1,000 items, reference labels shuffled *after* decoding). BLEU collapses from 78.6 to ∼1.0, confirming output–reference alignment. This test alone cannot distinguish pose-conditioned output from ID/cache-based retrieval; the input-side pose permutation below addresses that.

Condition	BLEU	EM
Original (correct pairing)	78.61	0.701
Permutation 0	1.06	0.000
Permutation 1	1.30	0.000
Permutation 2	0.90	0.002

Table 19: Input-side pose permutation with output-equivariance (100 items). Pose tensors are shuffled *before* decoding while IDs stay fixed. The equivariance column verifies $E(\text{pose}_j, \text{slot}_i) = E(\text{pose}_j, \text{slot}_j)$.

Condition	BLEU vs slot ref	Detail
Original decode vs own reference	75.62	baseline
Shuffled poses vs slot reference	0.95	BLEU collapses
Shuffled poses vs donor reference	75.62	identical to original
Output-equivariance match rate	—	100/100 (100%)
Donor-length match rate	—	100/100 (100%)
Slot-length match rate	—	5/100 (5%)

Table 20: Stratified exact-match rate on the full 7,060-item training pool under beam=3 free-decode.

Stratum	n	EM
By signer		
Signer01	1,852	0.683
Signer02	83	0.590
Signer03	579	0.736
Signer04	1,049	0.748
Signer05	1,623	0.701
Signer06	35	0.714
Signer07	762	0.696
Signer08	844	0.715
Signer09	233	0.734
By reference length		
Short (≤5 words)	203	0.837
Medium (6–10)	1,319	0.685
Long (11–20)	4,335	0.714
Very long (>20)	1,203	0.686
By text repetition in train		
Unique (1 occurrence)	6,777	0.702
Low (2–3)	57	0.737
Mid (4–10)	76	0.684
High (>10)	150	0.940

Table 21: Same-signer held-out control: train (seen) vs. test (unseen) free-decode. The 67-point BLEU gap is consistent with training-pool exposure contributing to the readout, though the lack of full covariate matching means we cannot quantify how much of the gap is exposure vs. distributional shift.

Split	n	BLEU	EM	Shared signers
Train (seen)	7,060	78.82	0.707	9
Test (unseen)	500	12.19	0.032	9 (same)